Tristan E. Listanco

Software Development | Embedded Systems & IoT | Applied Research

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Profile

Computer Applications student (Embedded Systems) at Mindanao State University – Iligan Institute of Technology, building software across embedded firmware, mobile, web, and computer vision. Lead student member of a university-approved research project on IoT-based aphid detection in bell pepper farms, with one paper at ICICT 2026 (London) as co-author and one at EUSPN 2026 (Almaty) as main author. Projects have placed in competitions run by UNESCO, Huawei, and the Swiss Innovation Prize.

Education

BS in Computer Applications, Major in Embedded Systems — Mindanao State University – Iligan Institute of Technology (2023–present)

- Coursework and focus: IoT systems, computer programming, software and application development, embedded systems, computer networking, database management, sensor and device integration, data processing, and systems design.

Science, Technology, Engineering and Mathematics (Senior High) — Iligan City National High School (2023), with High Honors

Awards and Recognition

- **Champion** — Analog Devices Inc. Academe Expo 2026, University Icon (U-ICON) (September 2026).
- **Philippine Delegate** — Huawei Seeds for the Future 2025 (September 2025).
- **National Winner, Food and Agribusiness** — DEPDev H.A.B.I. Workshop 2025 (April 2025).
- **Finalist** — Huawei Developer Cloud Competition, Asia Pacific Region, Student Category (December 2024).
- **Top 3 National Finalist** — Swiss Innovation Prize 2024, Digitalization Stream (November 2024).
- **Champion** — Sustainable Innovation Pitch Competition, SDG Day 2024 (September 2024).
- **Outstanding Award for Science and Technology Innovation** — ISPARK International Pitching Competition 2024, Tunku Abdul Rahman University of Management and Technology (August 2024).
- **Finalist** — UNESCO Water Resilience Challenge 2024 (Road to Bali) and 2023 (Youth Action for Water).
- **Gold Awardee** — Malaysia Virtual Innovation Competition 2023 (June 2023).
- **Finalist** — National Science and Technology Fair 2022, Robotics and Intelligent Machines (individual) (August 2022).
- **1st Runner Up and Best in UI Prototype** — Make Waves Hackathon and Pitching Competition (July 2021); **National Qualifier** — Impact 2050 Hackathon (August 2021).

Selected Software and Embedded Projects

- **Pepper Watch (iOS and watchOS)**, 2026 — Offline, on-device aphid detection for bell pepper fields in Swift using Create ML. Locates each leaf, classifies aphid damage, and keeps a per-field history; an Apple Watch app and widgets show field status. GitHub
- **Pepper Watch on Raspberry Pi 5**, 2026 — Camera-based crop monitoring running YOLOv8n on live frames. The Pi hosts its own Wi-Fi network and serves a live dashboard to any phone, with no internet, cloud account, or app install. GitHub
- **RT-Spark MP3 Player**, 2026 — Eight-song player for the RT-Thread RT-Spark board (STM32F407ZGT6), built with PlatformIO, STM32Cube HAL, and FreeRTOS without the Arduino framework. GitHub
- **FreeRTOS Multisensor Room Monitor**, 2026 — Room-monitoring device on the STM32F103C8 “Blue Pill” using STM32Cube HAL and native FreeRTOS, simulated in Wokwi. GitHub
- **Live Scoring Tabulations System**, 2026 — Web tabulation for judged competitions: an admin sets up an activity, judges score from their own devices, and a public page ranks results live. GitHub
- **Project AquaGuard**, 2024 — Android app in Kotlin for a water quality monitoring device. GitHub
- **Online Web LMS**, 2023 — Learning management progressive web app built on the MERN stack. GitHub
- **Practical Research for Beginners**, 2022 — React Native progressive web app guiding secondary students through research. GitHub

Research Experience and Publications

Publications

Listanco T.; Cañete P.; Tanque S. R.; Alce A. R. B.; Castor P. R. P. "Computational Approaches, Evidence Maturity, and Research Gaps in Bell-Pepper Disease and Pest Monitoring: A Systematic Literature Review." *17th International Conference on Emerging Ubiquitous Systems and Pervasive Networks (EUSPN 2026)*, Almaty, Kazakhstan, October 28–30, 2026. Main author. Accepted and presented. Paper ID 75.

Llantos O. E.; Hontiveros H. A.; **Listanco T. E.**; Lluch J. R. S.; Loable V. J. F.; Montefalcon P. A., II; Pesaras N. T.; Yancha C. T. D. "CLENS Alpha: A Mobile-Based Prototype for Digital Waste Pickup Coordination Using Real-Time Tracking and Confirmation Workflows." *Proceedings of the 11th International Congress on Information and Communication Technology (ICICT 2026)*, London, United Kingdom, February 24–27, 2026. Lecture Notes in Networks and Systems, vol. 2116, pp. 207–216. Springer, Cham. Co-author. DOI: 10.1007/978-3-032-36012-0_17

Research and Development Projects

- **Lead Student Member** — *Development of an IoT System for Aphids Damage Detection in Bell Pepper (Capsicum annuum) Farms of Northern Mindanao*, university-approved research project, College of Computer Studies, MSU-IIT (Jan–Dec 2026).
- **Member** — *CLENS: Garbage Awareness App using Gamification and IoT*, university-approved research project, College of Computer Studies (2025).
- **Member** — AquaGuard pilot-scale implementation in Iligan City, technological extension project, Department of Computer Applications (2025).
- **Founder** — *AquaGuard*: IoT-based monitoring system for communal water filtration and storage, funded by UNESCO Asia (2024–2026). Finalist at the UNESCO Water Resilience Challenge 2024 (Road to Bali), which granted USD 1,000 seed funding for a Water Based Innovation pilot in Puerto Princesa, Palawan.

Leadership and Professional Experience

- **College Vice Governor** — Student Council, College of Computer Studies, MSU-IIT (present). Represents college students before the administration and leads welfare and academic initiatives.
- **College Board Member for Multimedia Management** — Student Council, College of Computer Studies, MSU-IIT (2025–2026).
- **Focal Person** — UNESCO SeaBRNet Youth (Philippines) (2025–2026). Advocacy on environmental sustainability and water conservation.
- **Mobile Engineer** — Build My Wealth Inc. (January–September 2022).
- **President** — Electrobotz Club, Iligan City National High School (2022–2023).

Technical Skills

Languages — Embedded C/C++, AVR Assembly, Python, Swift, Kotlin, JavaScript.

Embedded and IoT — STM32 (Cube HAL), FreeRTOS, PlatformIO, Raspberry Pi 5, Wokwi simulation, MQTT, Adafruit IO, Blynk, wireless sensor networks.

Computer Vision and ML — YOLO (YOLOv8n) and OpenCV object detection; on-device models with Apple CreateML.

Mobile — Swift (iOS and watchOS), Android Jetpack and Kotlin, React Native, React.

Web and Data — Next.js with server-side rendering, MERN stack, progressive web apps, Supabase, MySQL.

Tools — Git, GitHub, GitLab; Figma for mobile and desktop prototyping; technical writing and code documentation.

Languages (spoken) — English, Filipino, Cebuano (proficient, verbal and written).